

5. Data Analysis

Data analysis is the process of examining, organizing, and analyzing data to find useful information and business insights.

In this project, data analysis was performed using Microsoft Excel to understand:

- Sales performance
- Customer behavior
- Order trends
- Top-performing states
- Best sales channels

The main goal of data analysis was to convert raw sales data into meaningful insights using PivotTables, PivotCharts, slicers, and dashboards.

Data Analysis Steps Used in the Project

1. Orders vs Sales Analysis

Steps Performed

1. Went to:

Insert → PivotTable

2. Added:

- Amount → Values box
- Order ID → Values box
- Month → Rows box

3. Went to:

PivotTable Analyze → PivotChart

4. Selected:

Combo Chart

5. Enabled:

Secondary Axis

6. Hid field buttons:

Right Click → Hide All Field Buttons

7. Added chart title:

Orders vs Sales

Purpose

Compared:

- Monthly sales amount
- Total orders

Benefit

Helped identify monthly sales trends and order performance.

2. Men vs Women Analysis

Steps Performed

1. Created new PivotTable.
2. Added:
 - Gender → Rows box
 - Amount → Values box
3. Inserted:

Pie Chart

4. Formatted chart and copied it to dashboard.

Purpose

Compared sales contribution by:

- Men
- Women

Benefit

Helped understand customer purchasing behavior.

3. Order Status Analysis

Steps Performed

1. Created new PivotTable.
2. Added:
 - Status → Rows box
 - Order ID → Values box
3. Inserted:

Pie Chart

4. Formatted and copied chart to dashboard.

Purpose

Analyzed:

- Delivered orders
- Cancelled orders
- Returned orders
- Refunded orders

Benefit

Helped track order performance and delivery success.

4. Top 5 States Analysis

Steps Performed

1. Created new PivotTable.
2. Added:
 - Ship State → Rows box
 - Amount → Values box
3. Sorted values:

Largest to Smallest

4. Applied:

Filter → Top 10 → Top 5

5. Inserted:

Bar Chart

Purpose

Identified states generating highest sales.

Benefit

Helped identify top-performing locations.

5. Age vs Gender Analysis**Steps Performed**

1. Created new PivotTable.
2. Added:
 - Age Group → Rows box
 - Gender → Columns box
 - Order ID → Values box
3. Used:

Show Values As → % of Grand Total

4. Inserted:

Column Chart

Purpose

Compared customer age groups with gender contribution.

Benefit

Helped identify major customer segments.

6. Channel Analysis**Steps Performed**

1. Created new PivotTable.
2. Added:
 - Channel → Rows box
 - Order ID → Values box
3. Used:

Show Values As → % of Grand Total

4. Inserted:

Pie Chart

Purpose

Compared sales contribution from:

- Amazon
- Myntra
- Flipkart
- Ajio

Benefit

Helped identify best-performing sales channels.

7. Dashboard Design

Steps Performed

- Added dashboard title
- Removed gridlines
- Formatted charts
- Adjusted layout properly
- Organized charts neatly

Features Used

Home → Merge & Center

View → Gridlines → Uncheck

Purpose

Created a professional and user-friendly dashboard.

Benefit

Improved dashboard readability and presentation.

8. Slicer Integration

Steps Performed

1. Selected PivotChart.
2. Went to:

PivotChart Analyze → Insert Slicer

3. Added slicers:
 - Month
 - Channel
 - Category
4. Right-clicked slicer.
5. Selected:

Report Connections

6. Connected slicers to all PivotTables.

Purpose

Allowed users to filter the entire dashboard interactively.

Benefit

Made dashboard dynamic and easy to use.

9. Dashboard Insights

Analysis Performed

- Monthly sales trends
- Gender-wise sales analysis
- Order status analysis
- State-wise sales analysis
- Age group analysis
- Channel contribution analysis

Insights Found

- Women contributed higher sales
- Adults were major buyers

- Delivered orders were highest
- Amazon and Myntra generated major sales
- Some states generated highest revenue

Benefit

Helped understand business performance and customer behavior for better decision-making.